Exact Hankel Periods, Parking-Outcome Formulas, and Three Further Investigations

KEIO

T96–T100 | 27 October 2026

Abstract

We prove Han's least-period conjecture for every shifted binary Rudin–Shapiro Hankel sequence with shift at least three. The proof combines explicit norm-one units in a quadratic function field with an approximation-error criterion: a sufficiently long gap between normal indices forces a square norm, and the possible finite pole orders then force that norm to be one. This identifies the exact spacing of long gaps and proves minimality, in addition to periodicity. We also give nonrecursive coefficient-extraction formulas answering two parking-outcome enumeration questions of Ferreri, Harris, Martinez, and Swartz, including arbitrary repeated capacities and every position of a single unavailable spot. Three further investigations concern Stern parity counts, rank-two average mixing, and a Legendre-determinant nonvanishing conjecture. Their incomplete conclusions are explicitly separated from the complete original-target proofs. Written arguments, primary-source comparisons, and exact reproducibility programs accompany the results. Worldwide priority and a historically unprecedented foundational principle are not certified.

Scope and success criteria

The five tracks below are research attempts, not five asserted solutions of open problems. The original targets are identified before the auxiliary constructions, and their parameter ranges and mathematical objects are retained. A complete solution, a substantive new argument, and historical priority are separate assertions.

The main complete proof is T98: Han's 2016 Conjecture 4.2, including the least-period requirement for every shift. T97 answers two explicit questions in Section 6 of the 2025 parking-outcome paper; those questions have author-and-section provenance but no individual conjecture numbers. The terminal-event and increasing-run decompositions used there extend classical methods. The three other tracks must be read with their stated remaining gaps; their auxiliary statements do not replace their original targets.

The requested combination of an unchanged previously open named problem, an unconditional complete proof, a historically unprecedented principle, and established absence of an identical prior resolution is not certified for all five tracks, or asserted as an absolute historical certification for any one track. Bounded literature searches are reported as such. Internal mathematical reviews and exact calculations support the proofs; they are not external peer review or proof-assistant formalizations.

This round follows the preceding T91–T95 investigation. Its previously proved AFM lucky-set identity and zero-residue Stern growth theorem are not counted again. The supplied mathematical corpus is treated as a collection of research drafts, consistent with its own provenance statement; original primary papers are used to establish the external targets and the published inputs.

1 T96: a degree-preserving ternary fold for odd Stern coefficients

The unchanged target is Question 2.10 of Ulas [1]: for $r \in \{0,1\}$, does $\Pi_{r,2}(x)/(\log_2 x)^2$ tend to infinity? Here

$$B_0 = 0, \quad B_1 = 1, \quad B_{2n} = tB_n, \quad B_{2n+1} = B_n + B_{n+1}, \qquad e(n) = \deg B_n,$$

and

$$\Pi_{r,2}(x) = \#\left\{1 \le n \le x : B_n(t) \equiv 1 + r\sum_{i=1}^{e(n)} t^i \pmod 2\right\}.$$

The degree in this definition is the degree over $\mathbb{Z}$, not the possibly smaller degree after reduction modulo two. The earlier T92 argument treated $r = 0$. The construction below advances $r = 1$, but its exponent is less than two, so it does not close the original question.

Theorem 1.1 (A lower bound in the residue-one branch)**.** *Put $\alpha = \log_2 3$. For $L = \lfloor \log_2 x \rfloor \ge 3$,*

$$\Pi_{1,2}(x) \ge 3^{\lfloor \log_2((L+2)/5)\rfloor} \ge \frac{1}{3}\left(\frac{L+2}{5}\right)^{\alpha}.$$

In particular $\Pi_{1,2}(x) \gg (\log x)^{\log_2 3}$, and $\Pi_{1,2}(x)/(\log x)^A \to \infty$ for every fixed $A < \log_2 3$.

The auxiliary problem is to find several reversible operations on binary words which preserve the full integer-degree congruence, rather than only its reduction. A parity identity gives three candidate operations; an integer leading-coefficient calculation proves that none creates hidden even coefficients in higher degrees.

For a binary word w, write $[w]_2$ for its value and $\mathrm{rev}(w)$ for its reversal. Products of the digit matrices

$$L = \begin{pmatrix} t & 0 \\ 1 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 1 \\ 0 & t \end{pmatrix}$$

act on $v = (0,1)^{\mathsf{T}}$ in the order in which digits are read. Thus $M_{ab} = M_b M_a$ and $M_w v = (B_{[w]_2}, B_{[w]_2+1})^{\mathsf{T}}$.

Lemma 1.2 (The ternary fold)**.** *Suppose that w has at least three digits, starts and ends in 11, and*

$$B_{[w]_2}(t) \equiv \sum_{i=0}^{E} t^i \pmod 2, \qquad E = e([w]_2).$$

Remove its last digit, writing $w = g1$. For $k = 0, 1, 2$, define

$$\Phi_k(w) = g\ \mathrm{rev}(g)\, 0^k 11.$$

Each resulting word starts and ends in 11, and

$$e([\Phi_k(w)]_2) = 2E + 1, \qquad B_{[\Phi_k(w)]_2}(t) \equiv \sum_{i=0}^{2E+1} t^i \pmod 2.$$

The three maps have disjoint images, and each has an explicit inverse on its image.

Proof. First work over $\mathbb{F}_2[t]$. Put $d = t + 1$ and $K = \left(\begin{smallmatrix} d & 1 \\ 1 & d \end{smallmatrix}\right)$. Direct multiplication gives $KL = L^{\mathsf{T}}K$ and $KR = R^{\mathsf{T}}K$. Hence

$$KM_{\mathrm{rev}(g)} = M_g^{\mathsf{T}} K.$$

Also $(1,0)R^2 = (1,d) = v^{\mathsf{T}}K$ and $(1,d)L^k = (1,d)$ for every $k \geq 0$. If $M_g v = (a,b)^{\mathsf{T}}$, then $B_{[w]_2} = a + b$, and therefore

$$\begin{aligned} B_{[\Phi_k(w)]_2} &= v^{\mathsf{T}} K M_{\mathrm{rev}(g)} M_g v \\ &= (a,b)K(a,b)^{\mathsf{T}} = d(a+b)^2 = \sum_{i=0}^{2E+1} t^i. \end{aligned}$$

The last equality uses $(1+t)(1+t+\cdots+t^E)^2 = 1+t+\cdots+t^{2E+1}$ in characteristic two. This parity calculation alone would not suffice.

We now bound the degrees before reduction. All digit-matrix entries have nonnegative integer coefficients. The degree recurrence implies $|e(n+1) - e(n)| \leq 1$ for $n \geq 1$, by induction. It also implies $e(n+1) \geq e(n)$ when n is odd. Since g ends in 1,

$$E = e([g1]_2) = e([g]_2 + 1).$$

Write $M_g = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$. Thus $\deg q \leq E$ and $\deg s = E$. Because g begins with 11, we can write $M_g = AR^2$, where A again has nonnegative coefficients. The second column of AR^2 equals d times its first column plus t^2 times the second column of A. Consequently

$$\deg p \leq E - 1, \qquad \deg r \leq E - 1.$$

We use the convention $\deg 0 = -\infty$ here.

Over $\mathbb{Z}[t]$ the exact self-adjointness matrix is

$$K_{\mathbb{Z}} = \begin{pmatrix} 1-t & 1 \\ 1 & 1-t \end{pmatrix}, \qquad K_{\mathbb{Z}} M_{\mathrm{rev}(g)} = M_g^{\mathsf{T}} K_{\mathbb{Z}}.$$

Its leading matrix is $-I$. Comparing the highest powers of t on the two sides shows that $M_{\mathrm{rev}(g)}$ has the same maximum degree E as M_g, with transposed leading-coefficient matrix. The first column of M_g has no term of degree E; the first row of $M_{\mathrm{rev}(g)}$ therefore has degree at most $E - 1$. It follows that, on putting

$$M_{\mathrm{rev}(g)} M_g v = (A_0, B_0)^{\mathsf{T}},$$

one has $\deg A_0 \leq 2E - 1$ and $\deg B_0 \leq 2E$.

Let $S_k = 1 + t + \cdots + t^{k-1}$, with $S_0 = 0$. Over the integers,

$$(1,0)R^2 L^k = (t^k + dS_k, d).$$

For $0 \leq k \leq 2$ the first entry has degree at most two. Thus

$$\deg B_{[\Phi_k(w)]_2} \leq \max(2E - 1 + k, 2E + 1) = 2E + 1.$$

The parity calculation already gives a nonzero coefficient in degree $2E + 1$, proving equality and the full required congruence.

Finally, $g\,\mathrm{rev}(g)$ ends in 1. For $k = 0$ the output ends in at least three ones; for $k = 1$ or 2, the final run 11 is preceded by exactly k zeros. These distinguish the three images. After deleting $0^k 11$, take the first half of the remaining word to recover g, and append 1 to recover w. This proves injectivity. □

Proof of the theorem. Start with $w_0 = 111$, so $B_7 = 1 + t + t^2$. For every string $(k_1, \ldots, k_j) \in \{0,1,2\}^j$, apply the corresponding folds. Their inverses give exactly 3^j distinct binary words at level j. All of them satisfy the original residue-one congruence, with

$$e([w_j]_2) = 3 \cdot 2^j - 1.$$

The lengths satisfy $\ell_0 = 3$ and $\ell_i = 2\ell_{i-1} + k_i$. In particular,

$$\ell_j = 3 \cdot 2^j + \sum_{i=1}^{j} 2^{j-i} k_i \le 5 \cdot 2^j - 2.$$

For $j = \lfloor \log_2((L+2)/5) \rfloor$, all these words have at most L digits, and hence represent indices below $2^L \le x$. This proves the first bound. The second uses $3^{\lfloor y \rfloor} \ge 3^{y-1}$. □

What is and is not resolved. The source proves the lower bound $\Pi_{1,2}(x) \ge \log_2 x$ in Corollary 2.7. The exponent $\log_2 3$ above is a stronger quantitative bound for exactly the same counting function, with its integer-degree condition intact. It is still smaller than the exponent two in Question 2.10, so this is a partial advance, not a complete answer. The preceding round's $r = 0$ result is not counted again as a new result.

Digit matrices, reversal symmetries, and characteristic-two quadratic identities are established tools. Related reversal work includes Spiegelhofer [2], for a different polynomial generalization of Stern's sequence. The specific contribution proved here is the three-branch operation together with the integer leading- degree argument needed for Ulas's exact condition. The bounded search did not locate this construction or its stated lower bound; that is not a certification of a historically unprecedented principle or worldwide priority. This track does not meet the requested full-success gate.

The program `t96_verify.py` checks the integer self-adjointness identities, the entire ternary construction through level six, additional original residue-one seeds, and independent integer polynomial examples. It includes negative controls showing why extending k to three or dropping the initial-11 restriction can introduce a degree gap.

2 T97: terminal-event enumeration and two parking-outcome formulas

The targets are the first two enumeration questions in Section 6 of Ferreri–Harris–Martinez–Swartz [3]. The first asks for the number of outcomes admitting k lucky cars for an arbitrary capacity vector, including repetitions. The second asks for a closed formula for $c_s(n, I)$, the number of outcomes with a fixed lucky set I when the single unavailable spot is s. These questions are explicit in the pre-existing source, although they are not assigned conjecture numbers. We give nonrecursive coefficient-extraction formulas over their full parameter ranges. The outcomes, rather than the preference lists, are counted, and an outcome admitted by several lucky sets is counted once.

The auxiliary device is to order the terminal occurrences of letters and enumerate the intervals between them. It yields a rational series with only $r!$ summands for r available spots, independent of their capacities. A separate increasing-run decomposition gives all values of $c_s(n, I)$ simultaneously from $2^{|I|}$ products. These are complete enumerative formulas, but the use of classical order decompositions and formal geometric series does not establish the requested historically unprecedented foundational principle.

2.1 An auxiliary terminal-event problem

Fix r letters, each required to occur at least once in a finite word. The final occurrence of letter j is its terminal event. Fix their order $\sigma = (\sigma(1), \dots, \sigma(r))$, and write $\rho(j)$ for the rank of j in this order. The hth interval, for $0 \le h < r$, consists of the ordinary occurrences after the first h terminal events and before the next terminal event. Only letters with $\rho(j) > h$ can occur in this interval. Give an ordinary occurrence of j in interval h weight $x_j w_{j,h}$, and its terminal occurrence weight a_j.

Lemma 2.1 (Terminal-event generating series)**.** *For a fixed terminal order, the generating series of these words is*

$$\frac{\prod_{j=1}^{r} a_j}{\prod_{h=0}^{r-1} \left(1 - \sum_{\rho(j)>h} x_j w_{j,h}\right)}.$$

Consequently the coefficient of $\prod_j x_j^{m_j-1}$ counts words with m_j occurrences of each letter, with the stated weights.

Proof. The word has a unique decomposition

$$W_0\, \sigma(1)\, W_1\, \sigma(2) \cdots W_{r-1}\, \sigma(r),$$

where W_h is any word on the letters whose terminal events have not yet occurred. The ordinary generating series of W_h is the geometric series of the sum of its allowed letter weights. Multiplying these series and the terminal weights proves the formula. Conversely every such choice has precisely the indicated terminal order: its terminal occurrence is present and the letter is forbidden afterward. Thus the decomposition is bijective, including empty intervals. □

2.2 Arbitrary capacities: the full outcome-counting formula

Write the distinct available spots as $v_1 < \cdots < v_r$, with positive capacities $m_1, \dots, m_r$ and $n = \sum_j m_j$. Thus the vector $\boldsymbol{u}$ contains m_j copies of v_j. Every outcome is an ordered set partition $B = (B_1, \dots, B_r)$ of $[n]$ with $|B_j| = m_j$. All such partitions are possible outcomes, by

letting each car prefer its designated spot. Define

$$F(B) = \bigcup_{v_j=1} B_j \ \cup \bigcup_{\substack{j\geq 2 \\ v_j=v_{j-1}+1}} \{i \in B_j : i < \max B_{j-1}\}.$$

This is the set of cars forced to be lucky by the outcome. The following criterion is the content of the source's Theorem 3.7; we include the short verification to specify precisely what is counted.

Lemma 2.2 (The possible lucky sets of one outcome)**.** *An outcome B admits the exact lucky set I if and only if $F(B) \subseteq I \subseteq [n]$.*

Proof. A car parking at spot 1 must prefer 1. If a car i parks at v_j while the preceding spot $v_j - 1 = v_{j-1}$ still has an unoccupied place, then every preference below v_j stops before v_j. Such a car must be lucky. The preceding spot is not full when i arrives exactly when $i < \max B_{j-1}$. This proves necessity.

For sufficiency, let each car in I prefer its designated spot v_j, and each car outside I prefer $v_j - 1$. The latter is positive: cars at spot 1 are in $F(B)$. The preceding spot is either unavailable or is already full, because its designated cars all have smaller labels. Induction on the arriving label shows that every car parks in its designated block and that the exact lucky set is I. □

Put

$$P_{\boldsymbol{u}}(q) = \sum_{B} q^{|F(B)|}.$$

This records the minimum possible number of lucky cars for each outcome, not the number for a single chosen preference list. For $\sigma \in \mathfrak{S}_r$ let $\rho_\sigma(j)$ be the rank of j in σ, and for $0 \leq h \leq r$ set

$$\epsilon_\sigma(j,h) = \begin{cases} 1, & v_j = 1, \\ 1, & j \geq 2,\ v_j = v_{j-1}+1,\ \rho_\sigma(j-1) > h, \\ 0, & \text{otherwise.} \end{cases}$$

Define the integer A_σ and linear forms $Q_{\sigma,h}$ by

$$A_\sigma = \sum_{j=1}^{r} \epsilon_\sigma(j, \rho_\sigma(j)), \qquad Q_{\sigma,h}(\mathbf{x}, q) = \sum_{\rho_\sigma(j) > h} x_j q^{\epsilon_\sigma(j,h)} \quad (0 \leq h < r).$$

Theorem 2.3 (General-capacity outcome formula)**.** *For every weakly increasing positive vector $\boldsymbol{u}$, with distinct spots and capacities as above,*

$$P_{\boldsymbol{u}}(q) = [x_1^{m_1-1} \cdots x_r^{m_r-1}] \sum_{\sigma \in \mathfrak{S}_r} \frac{q^{A_\sigma}}{\prod_{h=0}^{r-1}(1 - Q_{\sigma,h}(\mathbf{x}, q))}. \tag{1}$$

The number of distinct outcomes admitting exactly k lucky cars is

$$\left| \bigcup_{\substack{I \subseteq [n] \\ |I|=k}} \mathcal{O}_{\boldsymbol{u}}(I) \right| = [q^k] \frac{P_{\boldsymbol{u}}(q)}{1-q} \qquad (0 \leq k \leq n). \tag{2}$$

All series in (1) *are expanded formally at* $\mathbf{x} = 0$.

Proof. Encode an ordered partition B by the word $w_1 \cdots w_n$ where $w_i = j$ precisely when car i belongs to B_j. The terminal event of letter j occurs at the label $\max B_j$. Its relative order is a unique permutation σ.

Suppose an ordinary occurrence of j lies after h terminal events. It is forced lucky if $v_j = 1$, or if its preceding available spot is adjacent and that spot's maximum label still lies in the future. The latter condition is exactly $\rho_\sigma(j-1) > h$. Thus its weight is $x_j q^{\epsilon_\sigma(j,h)}$. At the terminal occurrence of j, the corresponding criterion is $\rho_\sigma(j-1) > \rho_\sigma(j)$; its weight is therefore $q^{\epsilon_\sigma(j,\rho_\sigma(j))}$. The auxiliary lemma now gives the summand for σ. Extracting the indicated capacities and summing the disjoint terminal orders proves (1).

An outcome with $|F(B)| = a$ admits a lucky set of size k precisely when $a \le k \le n$, by the preceding lemma. Hence its contribution to the union count is one for every $k \ge a$. Taking the indicated coefficient of $(1-q)^{-1}$ proves (2). In particular, we do not sum the cardinalities of the overlapping sets $\mathcal{O}_{\boldsymbol{u}}(I)$. □

Formula (1) is a finite, nonrecursive rational coefficient formula in r variables with $r!$ explicitly specified terms. It contains no sum over parking functions, outcomes, or choices of their n labels. For a fixed number r of available spots, truncation at the requested powers evaluates it using polynomially many arithmetic operations in n; the capacities may be arbitrarily large. No polynomial-time bound in variable r is asserted. When all capacities are one, the terminal orders are the outcome permutations themselves; that already-solved boundary case is not presented as an additional advance.

There is also a useful factorization. Split the available spots into maximal consecutive clusters, containing $N_1, \dots, N_b$ cars. Let $P_\ell(q)$ be the forced-lucky polynomial within the ℓth cluster, with its first spot forced only if that cluster contains spot 1. Then

$$P_{\boldsymbol{u}}(q) = \binom{n}{N_1, \dots, N_b} \prod_{\ell=1}^{b} P_\ell(q).$$

Indeed, choose the labels assigned to the clusters, then standardize their relative orders. The forced-lucky comparisons never cross an unavailable spot. This reduces the terminal-order sum to a separate sum for each consecutive cluster.

For example, for spots $1, 2$ with capacities a, b, a direct specialization gives

$$P_{(1^a,2^b)}(q) = q^a \sum_{j=0}^{b} \binom{a+j-1}{a-1} q^j.$$

Indeed, if $\max B_1 = a + j$, exactly j cars of B_2 precede it, and the other $a - 1$ cars of B_1 can be chosen in the displayed number of ways. For capacities $(3, 2, 2)$ at spots $(1, 2, 3)$ the formula gives

$$P_{\boldsymbol{u}}(q) = q^3 + 8q^4 + 61q^5 + 95q^6 + 45q^7.$$

Thus the numbers of outcomes admitting $k = 3, 4, 5, 6, 7$ lucky cars are $1, 9, 70, 165, 210$, respectively, with no duplication between lucky sets.

2.3 One unavailable spot and an arbitrary fixed lucky set

Now all spots in $[n+1] \setminus \{s\}$ have capacity one, and $I \subseteq [n]$ is fixed, with $k = |I|$. As in the source's Definition 3.14, let $c_s(n, I)$ count their possible outcomes with exact lucky set I. For $J \subseteq I$ and $j \notin I$ put

$$\lambda_J(j) = |J \cap [j-1]|, \qquad \mu_J(j) = 1 + |(I \setminus J) \cap [j-1]|.$$

Theorem 2.4 (Closed formula for every unavailable spot)**.** *For every $n \ge 1$, $1 \le s \le n+1$, and $I \subseteq [n]$,*

$$c_s(n, I) = [z^{s-1}] \sum_{J \subseteq I} |J|!\,(k - |J|)!\, z^{|J|} \prod_{j \in [n] \setminus I} (\lambda_J(j) z + \mu_J(j)). \tag{3}$$

Thus one polynomial with 2^k product summands provides all $n+1$ values, without recursion in s or enumeration of $n!$ outcomes.

Proof. The unavailable spot splits an outcome permutation into a left word of length $s-1$ and a right word of length $n-s+1$. Inside either word, every descent bottom must be in I. The first left letter, if present, must be in I; the first right letter need not be, because its preceding spot is unavailable. These conditions are exactly the forced-lucky criterion already proved.

Let J be the lucky labels on the left. Split the left word immediately before each label in J. Each resulting run starts with that lucky label and then contains only unlucky labels in increasing order, all larger than its first label. The $|J|$ runs have an arbitrary order. On the right, split immediately before each lucky label in $I \setminus J$. There is also a possibly empty initial run of unlucky labels. Regard this initial run as having a fictitious first label 0; it stays first, while the other $k-|J|$ runs have an arbitrary order.

For each unlucky label j, there are $\lambda_J(j)$ eligible left runs and $\mu_J(j)$ eligible right runs. Once its run is chosen, its position is forced by increasing order. Marking every left label by z gives the product in (3); the lucky labels on the left contribute $z^{|J|}$, and ordering the runs gives $|J|!(k-|J|)!$. Conversely, the outcome recovers all these runs, their order, and every assignment uniquely. Extracting the required left length proves the formula, including empty left or right words. □

At $s=1$ only $J=\varnothing$ contributes, recovering the source's Theorem 3.11:

$$c_1(n,I) = k! \prod_{j \notin I} (1 + |I \cap [j-1]|).$$

At $s=n+1$ only the all-left assignments contribute, recovering the classical formula $k! \prod_{j \notin I} |I \cap [j-1]|$. For $n=6$ and $I=\{1,3,5\}$, the full list for $s=1,\dots,7$ is

$$(144, 72, 82, 82, 82, 78, 36).$$

Scope, prior methods, and verification. These formulas answer the two stated enumeration questions with their original parameter ranges. In particular, general capacities are not replaced by the capacity-one case. The first formula is based on an auxiliary terminal-event enumeration and the source's established outcome criterion. The second extends the increasing-run argument of the source's Theorems 2.2 and 3.11. The precise formulas were not located in a bounded primary-source search through 27 October 2026; this does not certify worldwide priority. Neither terminal-order decomposition nor the increasing-run principle is claimed to be historically new. Accordingly, this track is not certified as meeting every strict novelty condition in the request.

The standard-library program `t97_verify.py` checks (1) against direct ordered partitions in 728 cases through $n=6$, covering every capacity composition and every adjacency pattern. It independently runs the original parking procedure for 125 capacity vectors through $n=4$, checking the entire family of possible lucky sets for each outcome. It also checks (3) in 1,792 fixed-set/spot cases through $n=7$. These finite checks supplement the bijective all-parameter proofs above.

3 T98: the exact shifted Rudin–Shapiro Hankel period

The unchanged target is Conjecture 4.2 of Han, *Hankel continued fraction and its applications*, *Advances in Mathematics* **303** (2016), 295–321 [4]. The conjecture concerns the binary sequence and determinants over $\mathbb{F}_2$, not the signed Rudin–Shapiro sequence over the integers. Let $u_n \in \mathbb{F}_2$ be the parity of the number of occurrences of 11 in the binary expansion of n. Set

$$f_k(x) = \sum_{n\geq 0} u_{n+k}x^n, \qquad h_k(n) = \det(u_{k+i+j})_{0\leq i,j<n}, \qquad h_k(0) = 1.$$

Theorem 3.1 (Han's original least-period formula)**.** *For every $k \geq 3$, let $m \geq 1$ be determined by $2^m < k \leq 2^{m+1}$. Then the sequence $(h_k(n))_{n\geq 0}$ is periodic from its first term, with least period*

$$R_k = 9 \cdot 2^m.$$

The auxiliary problem is to recognize unusually long gaps between normal Padé indices from the norm of an approximation error. In this problem, a square-congruence at one finite place forces the norm to be a square; odd pole orders at the only other possible finite pole then force that norm to be exactly one. This identifies the long gaps with explicit units. Their spacing proves *minimality* of the Hankel period, which does not follow from a continued-fraction period alone.

3.1 The quadratic field and the shifted lattices

Work throughout in characteristic two. Put $z = x^{-1}$, $y = z + 1$, and

$$F_k(z) = z^{-1}f_k(z^{-1}), \qquad P_k(z) = \sum_{i=0}^{k-1} u_i z^{k-1-i}.$$

Then $F_k = z^k F_0 + P_k$. Han's quadratic equation for f_0 is

$$x^3 + (1+x)^4 f_0 + (1+x)^5 f_0^2 = 0.$$

Consequently $g = yF_0$ satisfies

$$g^2 + g = y^{-3}.$$

The function field $K = \mathbb{F}_2(z, g)$ has involution $\overline{g} = g + 1$. In particular,

$$\mathrm{Tr}(F_k) = \frac{z^k}{y}, \qquad \mathrm{N}(F_0) = y^{-5}. \tag{4}$$

The chosen expansion of F_k at $z = \infty$ has strictly negative degree, whereas its conjugate has degree $k - 1$ for $k \geq 2$.

There is only one possible finite pole of F_0: the point Q above $y = 0$. In the variable $t = y^{-1}$ the curve is the smooth cubic $g^2 + g = t^3$. At its point at infinity,

$$v_Q(y) = 2, \qquad v_Q(g) = -3, \qquad v_Q(F_0) = -5.$$

Thus, if $A, B \in \mathbb{F}_2[z]$ and $A + BF_0$ has a finite pole, that pole is at Q and its order is one of $5, 3, 1$. Indeed, a negative valuation must equal $2\deg_y B - 5$; the polynomial A cannot cancel it.

Let P_+ and P_- be the points $(t, g) = (0, 0)$ and $(0, 1)$. They are the two places above $z = \infty$. The function

$$\varepsilon = \frac{g}{g+1} = 1 + y^3 g = 1 + y^4 F_0$$

has norm one, trace y^3, and divisor $3P_+ - 3P_-$. Its degree in the chosen Laurent expansion is -3. Every norm-one function with no finite pole is a power of ε. It also has no finite zero:

its norm is one, and a finite zero would require a finite pole of its conjugate. Thus its divisor must be $j(P_+ - P_-)$. Since the curve is a smooth cubic of genus one, $P_+ - P_-$ is not principal: a function with one simple pole would give a degree-one map to the projective line. Its order is exactly three because $3(P_+ - P_-)$ is the divisor just exhibited. Hence $3 \mid j$; division by the appropriate power of ε leaves a constant, and the only nonzero constant in $\mathbb{F}_2$ is one.

3.2 Units that preserve a shifted approximation lattice

Write $L_j(T) = \varepsilon^j + \varepsilon^{-j}$ when $T = \varepsilon + \varepsilon^{-1}$. These polynomials satisfy $L_0 = 0$, $L_1 = T$, and $L_{j+1} = TL_j + L_{j-1}$. For $j > 0$,

$$\operatorname{ord}_z L_j(y^3) = \begin{cases} 0, & 3 \nmid j, \\ 2^{\nu_2(j)+1}, & 3 \mid j, \end{cases} \qquad \operatorname{ord}_y L_j(y^3) = 3 \cdot 2^{\nu_2(j)}. \tag{5}$$

Here ν_2 is the exponent of two in j. For the first formula, $L_j(1) = 0$ precisely when $3 \mid j$, and $L_3(T) = T(T+1)^2$, while $y^3 + 1 = z(z^2 + z + 1)$. For odd a, $L_a(X) = X + O(X^3)$; use composition $L_a(L_b(T)) = L_{ab}(T)$ and the Frobenius identity $L_{2a} = L_a^2$. These facts prove both formulas.

Fix $k \geq 3$ and put $J = 3 \cdot 2^m$ with $2^m < k \leq 2^{m+1}$. For $U = \varepsilon^J$, define

$$b = \frac{yL_J(y^3)}{z^k}, \qquad a = U + bF_k.$$

By (5), b is a polynomial and is divisible by y^7. The function a is invariant under the involution, so belongs to $\mathbb{F}_2(z)$; it has no finite pole and is therefore a polynomial. Moreover $b \operatorname{Tr}(F_k)$ and $b \operatorname{N}(F_k)$ are polynomials. The latter norm has denominator dividing y^5, and b is divisible by y^7. Multiplication by U on

$$\mathcal{M}_k = \mathbb{F}_2[z] + \mathbb{F}_2[z]F_k$$

therefore has the polynomial matrix

$$\begin{pmatrix} a & b \operatorname{N}(F_k) \\ b & a + b \operatorname{Tr}(F_k) \end{pmatrix}. \tag{6}$$

Its determinant is $\operatorname{N}(U) = 1$, so it is unimodular and preserves coprime pairs of coefficients in both directions. The same argument applies to every power U^ℓ.

Conversely, if $\varepsilon^j = p + qF_k$ with polynomial p, q, taking traces forces

$$q = \frac{yL_j(y^3)}{z^k}.$$

Thus (5) implies $J \mid j$. This also proves that J is the least positive exponent of a unit in $\mathcal{M}_k$.

3.3 A period for all normal indices

We use the standard formal continued-fraction correspondence: the normal indices n, meaning $h_k(n) \neq 0$, are exactly the denominator degrees of the convergents to F_k, including the initial convergent $0/1$ of degree zero. For a reduced proper convergent p/q, with q monic and $\deg q = n$, write

$$\deg(qF_k + p) = -(n + d), \qquad d \geq 1.$$

Then d is the gap to the next normal index. Equivalently, a reduced fraction is a convergent precisely when $\deg(qF_k + p) < -\deg q$. These are the usual formal Padé/continued- fraction facts (the strict approximation criterion is Proposition 1 of van der Poorten [5]); Han's Hankel-fraction Theorem 2.1 [4] also identifies the normal indices as cumulative positive gaps. The nonzero determinants are all one because the coefficient field is $\mathbb{F}_2$.

Let $\alpha = p + qF_k$ represent such a convergent. Its conjugate has degree $n + k - 1$, by (4). Multiplication by U increases the vanishing order of α at P_+ by $R = 3J = 9 \cdot 2^m$, and increases the degree of its conjugate by R. By the unimodular matrix (6), the resulting $p' + q'F_k$ has coprime polynomial coefficients. Taking its trace shows $\deg q' = n + R$, while its degree is $-(n + R + d)$. It is therefore a convergent of degree $n + R$ with the same following gap d.

Conversely, multiply a convergent of denominator degree $n + R$ by U^{-1}, for $n \geq 0$. The same degree calculation gives a reduced convergent of degree n with its gap unchanged. Hence

$$h_k(n + R) = h_k(n) \qquad (n \geq 0).$$

This proves a period from the first term, including cases where some Hankel determinants vanish. It does not yet prove least period.

3.4 The long-gap norm criterion and least period

The norm argument solves the following auxiliary problem in an entire family of quadratic function fields.

Lemma 3.2 (Odd-pole norm rigidity)**.** *Let $s \geq 1$, $y = z + 1$, and $\phi = g/y$, where $g^2 + g = y^{-(2s-1)}$ over $\mathbb{F}_2(z)$. Let A, B be polynomials, not both zero, with $z^k \mid B$ and $k \geq 2s + 2$. If $\deg \mathrm{N}(A + B\phi) \leq 0$, then*

$$\mathrm{N}(A + B\phi) = 1.$$

Proof. Write

$$N = A^2 + \frac{AB}{y} + \frac{B^2}{y^{2s+1}} = \frac{V}{y^{2s+1}}, \qquad \deg V \leq 2s + 1.$$

At $z = 0$, $N \equiv A^2 \pmod{z^k}$, so N' vanishes to order at least $k - 1$. But

$$N' = \frac{yV' + V}{y^{2s+2}}, \qquad \deg(yV' + V) \leq 2s.$$

As $k - 1 > 2s$, this derivative is zero and N is a rational square. The odd-order pole of $y^{-(2s-1)}$ is totally ramified in the Artin–Schreier extension. Its unique point Q has $v_Q(y) = 2$, $v_Q(g) = -(2s - 1)$, and $v_Q(\phi) = -(2s + 1)$. If $A + B\phi$ had a finite pole, its valuation would therefore be a negative odd number $2 \operatorname{ord}_y B - (2s + 1)$. The norm has the same valuation at $y = 0$, since the residue degree is one, contradicting its being a square. There are no other possible finite poles. Hence N is a nonzero polynomial of degree at most zero, and so equals one. □

Lemma 3.3 (A long gap forces a norm-one error)**.** *Let $k \geq 6$, and let p/q be a reduced convergent of denominator degree n, with following normal-index gap $d \geq k - 1$. Then*

$$p + qF_k = \varepsilon^{\ell J}$$

for some positive integer ℓ. Conversely, each of these positive unit powers is such an approximation error. Its gap is exactly $k - 1$, and the next normal index is ℓR.

Proof. Put $\alpha = p + qF_k = A + BF_0$, where $A = p + qP_k$ and $B = qz^k$ are polynomials. Its norm is

$$N = \mathrm{N}(\alpha) = A^2 + \frac{AB}{y} + \frac{B^2}{y^5} = \frac{V(z)}{y^5}.$$

Its degree is $(n + k - 1) - (n + d) = k - 1 - d \leq 0$. Apply Lemma 3.2 with $s = 2$ to obtain $N = 1$. The proof of that lemma also shows that α has no finite pole, so the unit classification gives $\alpha = \varepsilon^j$; its positive vanishing order at P_+ gives $j > 0$. Polynomial membership gives $J \mid j$.

For the converse, the polynomial matrix for $\varepsilon^{\ell J}$ shows that its coefficients p, q in $\mathcal{M}_k$ are coprime. Taking traces gives

$$\deg q = \ell R - k + 1, \qquad \deg \varepsilon^{\ell J} = -\ell R.$$

Thus it is a reduced convergent, with gap $k-1$ and next denominator degree ℓR, as asserted. □

For $k \geq 6$, the endpoints of all gaps at least $k-1$ are now exactly $R, 2R, 3R, \ldots$. Every period of the binary Hankel sequence must preserve this set of gap endpoints sufficiently far from its initial index, so it must be divisible by R. Since R is already a period, it is the least period.

3.5 The three small shifts

For completeness, $k = 3, 4, 5$ admit short exact quadratic-transform certificates. Encode a polynomial $\sum a_i x^i \in \mathbb{F}_2[x]$ by the integer $\sum a_i 2^i$. Here the initial power-series tail is the original $f_k(x)$, not the Cauchy series $F_k(z)$. For a current tail $W(x)$ satisfying $A + BW + CW^2 = 0$ with $B = 1 + x^4$, let $d = \operatorname{ord}_x A$ and let D be the degree-$(d+1)$ truncation of x^d/W. Substitution $W = x^d/(D + x^{d+2}G)$ gives the exact update

$$(A, B, C) \longmapsto \left(\frac{AD^2 + x^d BD + x^{2d} C}{x^{2d+2}},\ B,\ x^2 A \right).$$

The following lists of pairs $(d+1, D)$ return exactly to their initial triple, so their cumulative first coordinates specify all normal indices.

k	Initial (A, C)	Cycle pairs $(d+1, D)$
3	$(1, 408)$	$(1,1),(2,1),(1,1),(1,3),(1,3),(1,1)$ $(1,3),(2,1),(1,3),(3,1),(2,7),(2,7)$
4	$(196, 816)$	$(3,1),(2,3),(1,1),(1,1),(3,1)$ $(1,3),(2,1),(1,1),(1,3),(3,15)$
5	$(98, 1632)$	$(2,1),(3,11),(2,5),(2,1),(2,3),(1,1)$ $(2,5),(1,3),(2,7),(4,1),(1,3),(2,5)$ $(4,1),(1,3),(1,1),(1,3),(1,1),(4,31)$

Their binary Hankel period words, beginning at $n = 0$, are respectively

3	110111111011001010
4	100101110011011100
5	10100101010110110100011010001111 1000.

Checking the proper divisors of their lengths shows that their least periods are $18, 18, 36$. This completes the all-k proof.

Provenance, novelty, and verification. The original conjecture, its coefficient field, and its least-period requirement are retained. The proof above supplies both the period and its minimality. The new proposed argument is the explicit shifted-lattice unit computation combined with the long-gap norm criterion. Quadratic function fields, formal continued fractions, units, and genus-one divisors are established tools; a historically unprecedented foundational principle is not certified here. In particular, the broad connection between norm-one units, large partial quotients, and continued-fraction quasi-periods is classical; see van der Poorten [5, Sections 2–3]. The characteristic-two odd-pole norm criterion and its application to the precise shifted lattice are the additional arguments used here. A bounded search located no prior resolution of this precise conjecture, but that is not a guarantee of worldwide priority. The accompanying verifier checks the finite cycles, unit matrices, trace orders, and independent small Hankel determinants. The infinite conclusion follows from the proof, not from these finite checks.

4 T99: exact rank-two average mixing on 128 and 256 vertices

The unchanged external target is Open Problem 4.1 in the final published version of Coutinho–Guo [6]: whether there are only finitely many graphs whose average mixing matrix has rank two. The numbering here is that of the published article, not its first arXiv version. *This investigation does not answer that finiteness question.* It constructs two explicit connected simple graphs of orders 128 and 256 and proves that their average mixing matrices have rank exactly two. The missing step is an unbounded construction, or a finiteness theorem.

For a real symmetric adjacency matrix A, write $A = \sum_\theta \theta E_\theta$ for its spectral decomposition. Its average mixing matrix is

$$\widehat{M} = \lim_{R\to\infty} \frac{1}{R}\int_0^R e^{itA} \circ e^{-itA}\, dt = \sum_\theta E_\theta \circ E_\theta,$$

where $\circ$ denotes entrywise multiplication. The final equality follows by integrating each exponential $e^{it(\theta-\tau)}$.

4.1 An auxiliary construction problem with an exact answer

Let $G = \mathbb{F}_2^d$, $N = 2^d$, and choose $S, T \subseteq G \setminus \{0\}$ and $B \subseteq G$. Define a graph $X(S, B, T)$ on $G \times \{0, 1\}$ by the edges

$$(g, 0) \sim (g + s, 0), \quad (g, 1) \sim (g + t, 1), \quad (g, 0) \sim (g + b, 1), \qquad s \in S,\ t \in T,\ b \in B.$$

Every group element is its own inverse, so this is an undirected simple graph. For $x \in G$, put $\chi_x(g) = (-1)^{x\cdot g}$ and

$$a_x = \sum_{s\in S} \chi_x(s), \qquad b_x = \sum_{b\in B} \chi_x(b), \qquad c_x = \sum_{t\in T} \chi_x(t),$$

$$t_x = a_x + c_x, \qquad u_x = a_x - c_x, \qquad D_x = u_x^2 + 4b_x^2.$$

The auxiliary question is to give an exact finite certificate, using only these integer sums, for simple adjacency spectrum and rank-two mixing.

Proposition 4.1 (Parity certificate and exact mixing formula)**.** *Suppose $|S|$ is even, $|T|$ and $|B|$ are odd, and the N ordered pairs (t_x, D_x) are distinct. Then $A(X(S, B, T))$ has $2N$ distinct eigenvalues, and*

$$\widehat{M} = \frac{J_{2N}}{2N} + \frac{\kappa}{2N} ss^{\mathsf{T}}, \qquad s = (\mathbf{1}_N, -\mathbf{1}_N)^{\mathsf{T}}, \qquad \kappa = \frac{1}{N}\sum_{x\in G} \frac{u_x^2}{D_x}. \tag{7}$$

Here $0 < \kappa < 1$, and $\operatorname{rank} \widehat{M} = 2$. If S spans G, the graph is connected.

Proof. For each x, the two-dimensional subspace consisting of χ_x on either fiber is invariant under adjacency. In its normalized character basis adjacency acts as

$$H_x = \begin{pmatrix} a_x & b_x \\ b_x & c_x \end{pmatrix}, \qquad p_x(z) = \det(zI - H_x) = z^2 - t_x z + a_x c_x - b_x^2.$$

All N such subspaces are mutually orthogonal and together exhaust the vertex space. This is the standard character decomposition of a semi-Cayley graph [7].

Since a sum of signs has the parity of its number of summands, u_x and b_x are odd for every x. Thus $D_x \equiv 1 + 4 \equiv 5 \pmod 8$. In particular $D_x > 0$ is not a square, and p_x is irreducible over $\mathbb{Q}$. Distinct pairs (t_x, D_x) give distinct monic polynomials p_x. Two distinct monic irreducible

quadratics over $\mathbb{Q}$ cannot have a common root. Their roots $(t_x \pm \sqrt{D_x})/2$ therefore give $2N$ distinct eigenvalues.

Put

$$K_x = 2H_x - t_x I = \begin{pmatrix} u_x & 2b_x \\ 2b_x & -u_x \end{pmatrix}.$$

Then $K_x^2 = D_x I$, so the two spectral projectors in this block are $P_{x,\pm} = (I \pm K_x/\sqrt{D_x})/2$. For fiber-first ordering of the vertices the full projector is $P_{x,\pm} \otimes (\chi_x \chi_x^{\mathsf{T}}/N)$. Since every entry of χ_x has square one,

$$\widehat{M} = \frac{1}{N^2} \sum_x (P_{x,+} \circ P_{x,+} + P_{x,-} \circ P_{x,-}) \otimes J_N.$$

The diagonal entries of the parenthesized 2×2 matrix are $(1 + u_x^2/D_x)/2$; its off-diagonal entries are $2b_x^2/D_x = (1 - u_x^2/D_x)/2$. Summing proves (7). The nonzero orthogonal vectors $\mathbf{1}_{2N}$ and s have eigenvalues 1 and κ, respectively; all vectors orthogonal to both have eigenvalue zero. Oddness makes both u_x and b_x nonzero, whence $0 < u_x^2/D_x < 1$ and $0 < \kappa < 1$. This proves the rank assertion.

If S spans G, the graph induced on fiber zero is connected. Oddness makes B nonempty, and any fixed $b \in B$ joins every vertex of fiber one to a vertex of fiber zero. The full graph is connected. □

4.2 Two explicit graphs and their finite proof certificates

Identify G with the integers $0, \ldots, 2^d - 1$ using binary coordinates, so addition is bitwise XOR. For a nonnegative integer $h < 2^N$, define

$$\mathcal{S}(h) = \{g \in \{0, \ldots, N-1\} : \lfloor h/2^g \rfloor \equiv 1 \pmod 2\}.$$

The following table gives the hexadecimal integers h_S, h_B, h_T; the sets are exactly $S = \mathcal{S}(h_S)$, $B = \mathcal{S}(h_B)$, $T = \mathcal{S}(h_T)$. Thus the specification is complete and does not depend on a random seed or on reproducing the search.

d	Set	Hexadecimal membership mask
6	S	`1d39f52fe7384174`
6	B	`37ec8fd2075e6984`
6	T	`29e5fdffe09a47da`
7	S	`623a7e9d385f8df96b33f5d522a56aa8`
7	B	`db196b1a5d26bcf475c59b8a07829823`
7	T	`c8975afdc8d54101d50561d3475eab7e`

Proposition 4.2 (Two finite examples)**.** *The displayed masks define connected simple graphs on 128 and 256 vertices, respectively, with simple adjacency spectrum and average mixing rank exactly two.*

Exact finite verification. The accompanying dependency-free program `t99_verify.py` decodes the six integers above. For $d = 6$, it gives $(|S|, |B|, |T|) = (34, 33, 39)$; for $d = 7$, it gives $(70, 63, 65)$. In both cases $0 \notin S \cup T$ and Gaussian elimination over $\mathbb{F}_2$ gives $\dim\langle S \rangle = d$. For each of the 64 or 128 characters the program computes a_x, b_x, c_x in two ways: direct signed sums and an integer Walsh transform. These agree exactly. It then checks that $D_x \equiv 5 \pmod 8$ and that every pair (t_x, D_x) is distinct. These are finite integer assertions, with no floating-point eigenvalue calculation. The program also independently constructs all edges and checks the graph's simplicity and connectivity. All assertions pass. Proposition 4.1 supplies the conclusion. The file `t99_verify.json` records the expanded sets, every quadratic polynomial's trace, determinant and discriminant, and the exact rational value of κ for independent inspection. □

The two fiber degrees are respectively $(67, 72)$ and $(133, 128)$. The construction and verifier certify these two graphs individually; neither numerical rank estimation nor an extrapolation from small orders is used in the proof.

4.3 The obstruction to the attempted infinite construction

One attempted recursion transformed a column vector of three Walsh coefficients by either

$$M_+ = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad M_- = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

It cannot yield a collision-free family at every depth. Applying the operations in either of the two orders $-+++--$ and $+---++$ gives exactly the same matrix product,

$$\begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Consequently two distinct branches give identical coefficient triples after six steps, for every initial vector. This rejects that specific recursion; it is not an obstruction to all possible constructions.

The spectral decomposition used above is classical, and the rank bound also has a general explanation: with simple spectrum, a graph automorphism sends each real normalized eigenvector to its positive or negative. Squared eigenvector entries are therefore constant on each automorphism orbit. If a subgroup has two orbits, the squared-eigenvector matrix has rank at most two, and so does its Gram matrix $\widehat{M}$. The average-mixing framework is also established in Coutinho–Godsil–Guo–Zhan [8].

Accordingly, the contribution here is the explicit finite construction, its parity certificate and exact mixing formula, together with a checkable failure of the proposed recurrence. A bounded literature search did not locate the identical examples or an answer to the published finiteness question. That search does not establish worldwide priority. No historically new foundational principle, infinite family, or complete solution of Open Problem 4.1 is asserted.

5 T100: the nonvanishing gap in Sun's quadratic-residue determinant

The unchanged target is Conjecture 4.10(ii) of Sun [9]. For a prime $p \equiv 3 \pmod 4$, put $n = (p-1)/2$ and write χ for the Legendre symbol, with $\chi(0) = 0$. The conjecture is

$$\chi(2 \det A_p) = 1, \qquad A_p = \left[\chi(j+k) + \chi(j^2+k^2)\right]_{0 \le j,k \le n}.$$

The factor 2 is outside the determinant. All indices and the prime range in this statement are retained.

Jiang–Sun [10, Theorem 1.1 and Remark 1.1] already prove that the displayed Legendre symbol is 0 or 1, and explicitly state that they cannot prove $p \nmid \det A_p$. Thus merely reproducing a square factorization would not resolve the conjecture. The present investigation also does *not* prove this nonvanishing. We give the precise surviving obstruction and an all-prime reason why a proposed extension to arbitrary half-systems cannot supply it.

5.1 The exact existing reduction

All calculations in this subsection take place over $\mathbb{F}_p$. Let

$$P(X) = \prod_{k=0}^{n} (X-k), \qquad r_m(X) = X^m \bmod P(X), \qquad q = (n+1)/2.$$

Define the $q \times q$ matrix

$$C_{ab} = \mathbf{1}_{a+b=q-1} \binom{n}{2b+1} + \binom{n}{a} [X^{2b+1}] r_{2n-2a}(X), \qquad 0 \le a, b < q.$$

The parity-block argument in Jiang–Sun gives

$$\det A_p = (-1)^q (\det V)^2 (\det C)^2, \qquad V = (j^u)_{0 \le j,u \le n}. \tag{8}$$

For clarity, Euler's criterion replaces the entries by $(j+k)^n + (j^2+k^2)^n$. Reducing each variable modulo P produces a bivariate polynomial of degree at most n in each variable. Its coefficient matrix has zero odd–odd block: the first summand has odd total degree, and in each term of the second summand at least one exponent is even and at most $n-1$. Ordering even exponents before odd ones gives a symmetric block matrix of the form

$$\begin{pmatrix} B & C \\ C^T & 0 \end{pmatrix}.$$

Multiplication by V and V^T gives the evaluated matrix, proving (8). This is the published reduction, credited here rather than claimed as new.

Since the elements $0, 1, \ldots, n$ are distinct, $\det V \neq 0$. Also $\chi(2)(-1)^q = 1$. Consequently the original missing assertion is *exactly*

$$\det C \neq 0 \quad \text{in } \mathbb{F}_p.$$

This is a reduction of the original target, not a proved replacement for it.

5.2 An auxiliary question and an obstruction to a general principle

A half-system is a set $H = \{0, h_1, \dots, h_n\} \subset \mathbb{F}_p$ containing exactly one element of every nonzero pair $\{a, -a\}$. For such a set define

$$P_H(X) = \prod_{h \in H} (X - h) = E_H(X^2) + XO_H(X^2), \qquad A_H = \left[\chi(h+k) + \chi(h^2+k^2)\right]_{h,k \in H}.$$

The proposed auxiliary question was whether the common norm identity for E_H, O_H, together with the half-system property, forces $\det A_H \neq 0$. It has the following complete negative answer.

Proposition 5.1 (A universal obstruction for half-systems)**.** *For every prime $p \equiv 3 \pmod 4$ and every half-system H,*

$$E_H(Z)^2 - ZO_H(Z)^2 = Z^{n+1} - Z. \tag{9}$$

Nevertheless, for every such prime there is a half-system for which A_H has an identically zero row and hence zero determinant even over the integers. One may take H to be zero together with the quadratic nonresidues modulo p.

Proof. The degree $n+1$ is even. The roots of $P_H(X)P_H(-X)$ are every nonzero field element once and zero twice, and its leading coefficient is one. Hence

$$P_H(X)P_H(-X) = X \prod_{a \in \mathbb{F}_p} (X - a) = X^{p+1} - X^2.$$

Taking even and odd parts gives (9).

Since $\chi(-1) = -1$, exactly one member of each pair $\{a, -a\}$ is a quadratic nonresidue. The proposed H is therefore a half-system. For each nonzero $k \in H$, the entry in row zero is

$$\chi(k) + \chi(k^2) = -1 + 1 = 0.$$

The entry at $(0,0)$ is zero as well. In this example $P_H(X) = X(X^n + 1)$, so $E_H(Z) = Z^q$ and $O_H(Z) = 1$ explicitly satisfy the same norm identity. Thus that identity does not prevent singularity. □

This is a counterexample to the auxiliary extension, *not* a counterexample to Sun's original conjecture. Sun specifies the consecutive half-system $H_0 = \{0, 1, \dots, n\}$, and replacing it by the nonresidues would change the problem.

The consecutive set supplies additional identities that survive the obstruction above:

$$P(-X - 1/2) = P(X), \qquad (X + 1/2)P(X + 1) = (X + 1)P(X). \tag{10}$$

Indeed, $n = -1/2$ in $\mathbb{F}_p$, reflection $k \mapsto n - k$ permutes the roots, and the second identity follows by shifting the product defining P. The first identity also says that P is a polynomial in $(X + 1/4)^2$. These interval identities distinguish the required set from an arbitrary half-system, but no triangular factorization or other nonvanishing proof for C was obtained from them.

Verification and success status. The exact verifier `t100_verify.py` checks the original matrix, the coefficient matrix, (8), both interval identities, the norm identity, and the nonresidue obstruction for all primes $p < 200$ with $p \equiv 3 \pmod 4$. All original determinants in that finite range are nonzero modulo p. This is finite evidence and does not establish the all-prime assertion. T100 remains an unsuccessful attempt at the unchanged named problem. The auxiliary negative result and the existing compression formula are not counted as a new complete solution or as a historically unprecedented method.

Original-target status and reproducibility

The scope of each conclusion is recorded below. "Complete" refers to the written proof of the exact source question, not to a certification of historical priority or to external acceptance.

Track	Pre-existing target	Conclusion of this manuscript
T96	Ulas, Question 2.10	Partial: residue-one lower bound with exponent $\log_2 3 < 2$; the requested limit is not proved.
T97	Ferreri–Harris–Martinez–Swartz, Section 6, first two questions	Complete coefficient formulas over the original parameter ranges. These questions have no individual conjecture numbers.
T98	Han, Conjecture 4.2	Complete all-shift proof of the exact least period, using the odd-pole norm lemma and explicit units.
T99	Coutinho–Guo, Open Problem 4.1	Exact finite rank-two examples; the finite-versus-infinite question remains unresolved.
T100	Sun, Conjecture 4.10(ii)	An auxiliary obstruction is proved; nonvanishing for the prescribed consecutive set remains unresolved.

The main distinction in T98 is the additional characteristic-two argument in Lemma 3.2, followed by the precise classification of long gaps for the shifted lattices. Its use of classical function-field and continued-fraction theory is explicit. T97 uses classical combinatorial decompositions. Neither track is described as a historically unprecedented foundational principle.

Literature scope. The source and priority review used exact paper identifiers, author and conjecture numbers, distinctive formulas, and related primary papers, through 27 October 2026. In particular, the shifted binary Rudin–Shapiro problem was separated from signed Rudin–Shapiro nonsingularity, general automaticity of Hankel arrays, and other quadratic-series periodicity results. The parking counts were separated from counts of preference lists. The average-mixing target was kept as the ordinary Cesaro average, and the nonzero requirement in Sun's conjecture was separated from the published square-or-zero result. No identical resolution of the two T97 questions or Han's precise T98 conjecture was located in that bounded review. This is negative search evidence, not a proof that no such literature exists. An unretrieved potentially related thesis lead in the parking search was not treated as cleared. The verification archive includes the detailed source audit.

Reproduction. The archive contains this standalone LaTeX source, the compiled PDF, five exact-arithmetic Python verifiers, their JSON results, a single runner, and a file-hash manifest. The programs use Python's standard library and do not require floating-point eigenvalue tests, symbolic algebra packages, or network access. Run

```
python3 run_all.py
```

from the extracted archive, without Python's `-O` option. The runner checks file hashes before execution and reports each program's exit status. Each individual verifier can also be run directly. The LaTeX file compiles with pdfLaTeX using standard AMS, Latin Modern, geometry, microtype, and hyperref packages.

Finite computations check the displayed certificates and independent instances. They do not prove the unbounded parameter assertions by themselves. The written bijections, degree arguments, function-field proof, and explicit counterexample construction supply the infinite conclusions stated in the respective sections. Internal independent mathematical reviews found no unresolved gap in those claimed proofs; this is not external peer review or proof-assistant certification.

Strict success condition. The demand that all five tracks satisfy every historical and mathematical condition is not met. T98 supplies a complete proof of an unchanged, pre-existing named conjecture as stated in its original source; T97 supplies two complete formulas for explicit source questions. The remaining three original targets are not declared solved. Worldwide priority and a historically unprecedented principle remain unverified, including for the complete proofs.

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